Alice Blanche Ekane

Senior AWS Infrastructure Engineer/ Site Reliability Engineer (SRE)

Rockville, MD,20855 | (301) 768-3274 | aliceekane04@gmail.com | LinkedIn |GitHub

Senior AWS Infrastructure Engineer / Site Reliability Engineer with 6+ years of experience designing, automating, and securing AWS and Azure environments for healthcare, enterprise, and managed service platforms. Led the architecture of production AWS solutions that improved system visibility, cut incident response time, and enabled cloud cost savings of over $120K annually. Automated infrastructure delivery with Terraform and CI/CD pipelines, reducing manual deployment errors by 40% and boosting provisioning speeds by over 30%. Eager to leverage deep cloud, automation, and security expertise to drive highly reliable, cost-efficient, and secure infrastructure solutions for your organization.

Skills:

Cloud Platforms: AWS (EC2, VPC, S3, IAM, CloudWatch, CloudTrail, Lambda, SQS/SNS, Config, GuardDuty, Security Hub, CloudFront, ALB/NLB, RDS/Aurora, Systems Manager, AWS Backup, EKS, AWS Organizations, WAF & Shield, Cost Explorer & Trusted Advisor, Secrets Manager, KMS, Inspector) Azure Exposure (Virtual Machines, Blob Storage, VNets, Azure AD, Azure Monitor, AKS, Azure DevOps)
Infrastructure & Systems: Infrastructure Provisioning, Server Administration, Linux, Windows, Active Directory, Azure AD, Group Policies, Endpoint Management, Patching, Backup & Recovery
IaC & Automation: Terraform, CloudFormation, Ansible, Bash, PowerShell, Azure DevOps Pipelines
DevOps & CI/CD: Jenkins, GitHub, GitLab, Git, Azure DevOps, CI/CD Pipelines, Docker, Kubernetes, AKS, DevOps Practices
Programming & Frameworks: Python (Boto3 for AWS automation: XML, JSON), Django, Django REST Framework, FastAPI, Fullstack Frameworks, Pandas, Data Visualization
AI/ML & Emerging Tech: Deep Learning, Large Language Models (LLM), OpenAI SDK
Security & Compliance: SIEM Monitoring, Vulnerability Management, Threat Detection, Log Analysis, Access Management, Identity Hardening
Networking: TCP/IP, DNS, DHCP, NAT, Subnets, Routing, VPN, Load Balancing, Firewall Configurations, Elastic IP Management, VNet Connectivity
Monitoring & Optimization: CloudWatch Metrics/Logs, CloudTrail, AWS Config, Log Insights, Resource Optimization, Cost Management, Cost Optimization
IT Operations: Incident Response, Troubleshooting, Root-Cause Analysis (RCA), Change & Incident Management, ServiceNow, Jira, SLAs/OLAs

Professional Experience

Senior AWS Infrastructure Engineer/Site Reliability Engineer (SRE) , CVS Health **Mar 2024 - Present | TX**

- Architected production AWS environments using EC2 and VPC while integrating CloudWatch Metrics/Logs, improving visibility and reducing incident response efforts by 6 hours per week.
- Engineered secure object storage workflows with S3 and implemented AWS Backup, reducing recovery time by 32% and lowering annual storage costs by $120K across 3 PB of data.
- Utilized Azure Blob Storage for backup and archival workflows, improving data accessibility and supporting secure retention policies for enterprise applications.
- Automated infrastructure provisioning with Terraform and implemented GitHub workflows, reducing manual deployment errors by 40% and accelerating provisioning by 32% across 50+ environments.
- Collaborated with DevOps teams using Azure DevOps and GitHub Actions pipelines, accelerating deployment cycles and improving CI/CD workflow reliability by 28%.
- Orchestrated identity and access governance using IAM and strengthened Security Hub monitoring, decreasing unauthorized access alerts by 50% and avoiding potential remediation costs of $50K per year.
- Assisted in configuring Azure AD integrated identity workflows alongside AWS IAM, strengthening access governance and improving authentication reliability across enterprise applications.
- Configured hybrid network setups with ALB/NLB and VPN, enhancing application uptime to 99.5% and reducing latency by 13%, supporting 12,000+ daily requests.
- Supported hybrid cloud initiatives leveraging Azure Virtual Machines, VNets, and Azure Monitor, improving cross cloud operational visibility across enterprise environments and reducing monitoring gaps by 24%.
- Implemented endpoint management and incident-response workflows using ServiceNow and Jira, cutting average resolution time by 32% and sustaining 99.9% system availability
- Implemented Amazon SNS and SQS to enable asynchronous messaging between distributed services, improving workflow reliability and reducing processing delays by 35%.
- Configured AWS Elastic IPs for production EC2 and VPN gateways, ensuring consistent endpoint connectivity and reducing DNS propagation delays by 40%.
- Managed hybrid cloud connectivity for production workloads, improving secure access reliability for 12,000+ daily healthcare transactions.

- Drove AWS resource optimization initiatives that cut annual cloud spend by 18% by identifying underutilized workloads and implementing right sizing initiatives.
- Streamlined DNS configurations within hybrid AWS environments, reducing propagation delays by 40% and improving secure application accessibility across healthcare environments.
- Automated AWS resource provisioning and monitoring with Python Boto3 (XML, JSON), reducing manual intervention by 42%.
- Built APIs using Django REST Framework and FastAPI, accelerating integration workflows by 38%.
- Developed secure fullstack Django applications for cloud metrics visualization, improving compliance reporting speed by 33%.
- Applied Pandas and visualization libraries to analyze AWS usage, increasing cost forecasting accuracy by 29%.
- Applied Deep Learning, Large Language Models (LLMs), and OpenAI SDK capabilities to enhance anomaly detection efficiency by 26%.
- Administered Linux based EC2 environments and automated configuration management with Ansible, reducing manual server maintenance efforts by 45% across production workloads.
- Implemented AWS WAF & Shield, decreasing malicious traffic incidents by 41%.
- Leveraged AWS Organizations for governance, cutting policy drift by 36%.
- Automated *Linux patching*, configuration drift remediation, and application deployments using Ansible playbooks, improving operational consistency and reducing deployment failures by 30%.
- Leveraged Ansible alongside Terraform to automate AWS resource orchestration, accelerating infrastructure deployments by 35% across 50+ environments.
- Contributed to AKS and Kubernetes operational support activities, improving container orchestration efficiency and deployment consistency across hybrid cloud workloads.
- Applied AWS KMS encryption controls, improving secure data exchange reliability by 32%.
- Performed Linux system administration including troubleshooting, package management, user access management, and performance tuning, resulting in reliable and high availability of healthcare applications.

AWS Infrastructure Engineer, Dell Technologies **Nov 2021 - Feb 2024 | TX**

- Provisioned cloud infrastructure with Terraform and integrated CI/CD pipelines, cutting deployment time by 15 hours per month and increasing release frequency to 20+ deployments.
- Integrated Azure DevOps pipelines with Terraform and CI/CD workflows, reducing deployment delays and improving automation reliability by 30%.
- Implemented directory services using Active Directory and Group Policies, reducing configuration errors by 30% and improving login success for 5,000+ users, avoiding approximately $25K per year in support overhead.
- Assisted with Azure AD administration and identity governance, improving authentication efficiency and reducing user access issues across enterprise systems.
- Monitored cloud and database services with AWS Config and Log Insights, detecting 95% of anomalies proactively and reducing incident escalations by 26%.
- Monitored hybrid cloud infrastructure using Azure Monitor and AWS Config, improving anomaly detection accuracy and reducing incident escalation rates by 26%.
- Enforced vulnerability mitigation using Vulnerability Management, decreasing critical risk exposure by 40%, preventing potential downtime costs across 100+ workloads.
- Managed enterprise Linux and Windows server environments, improving patch coverage by 41% and reducing system downtime by 18% across 2,500+ servers.
- Configured networking components including Subnets, Routing, and load balancing, improving application availability and reducing latency by 12% while strengthening SLA adherence and saving approximately $30K annually in penalties.
- Supported Azure infrastructure services including Azure Virtual Machines, VNets, Blob Storage, and Azure Monitor by automating deployment processes, which improved operational efficiency and expanded monitoring coverage across the environment.
- Used SNS with SQS subscriptions to decouple application components, increasing fault tolerance and improving message delivery success rates by 28%.
- Administered Linux-based AWS environments using AWS Systems Manager and CloudWatch, troubleshooting issues, applying patches, managing user access, and optimizing performance for over 2,500 enterprise servers, which maintained high system reliability and reduced downtime.
- Provisioned static IPs for NAT gateways and directory services, improving application availability, reducing latency by 12%, and saving approximately $30K annually in SLA penalties.
- Implemented proactive cost management strategies across multi account AWS environments, reducing operational expenses by 22% while maintaining SLA compliance.
- Enhanced DNS routing and subnet management to improve application availability, reducing latency by 15% and strengthening enterprise connectivity.
- Automated provisioning pipelines with Python Boto3, reducing manual configuration errors by 37%.

- Designed APIs with Django REST Framework and FastAPI, improving interoperability by 34%.
- Built Django fullstack applications for visualization, reducing troubleshooting time by 28%.
- Leveraged Pandas for cloud spend analysis, improving *optimization* accuracy by 31%.
- Researched Deep Learning and Large Language Models (LLMs) with OpenAI SDK capabilities, reducing incident escalations by 22%.
- Deployed and managed EKS clusters across multiple environments, improving release velocity by 36% and enabling faster, more reliable deployments.
- Collaborated on AKS based container deployments integrated with Kubernetes orchestration workflows, improving deployment consistency and operational scalability.
- Configured AWS WAF & Shield, reducing downtime incidents by 33%.
- Implemented standardized policies and automated controls in AWS Organizations, improving compliance reporting accuracy by 27%.
- Automated Linux server configuration, patching, and application deployments using Ansible, reducing manual operational efforts by 38% across enterprise workloads.
- Automated credential management with *AWS* Secrets Manager, reducing rollback incidents by 25%.
- Applied *AWS* KMS encryption controls, ensuring secure communication and reducing encryption related errors by 31%.
- Integrated Ansible automation with Terraform and CI/CD pipelines to streamline infrastructure provisioning and improve deployment consistency by 34%.

Cloud Operations Engineer, DXC Technology **Aug 2019 - Oct 2021 | VA**

- Supported operational workflows through structured incident response using ServiceNow, resolved 95% of incidents within SLA and kept system availability at 98.7%, saving 8 hours per week in manual recovery
- Resolved infrastructure disruptions through troubleshooting with CloudWatch and AWS Config, cut escalations by 25% and boosted end-user satisfaction for over 10 client accounts, avoiding $20K per year in support costs.
- Investigated recurring technical patterns using Root-Cause Analysis (RCA), mitigating 37% of repeat incidents and increasing system reliability.
- Monitored security posture with SIEM Monitoring and Log Analysis, detecting 85% of abnormal activities proactively and reducing potential breach costs by $45K/year.
- Facilitated change workflows using Jira, decreasing failed deployments by 18% and improving update compliance for 50+ components, avoiding unnecessary downtime costs.
- Configured secure connectivity layers via VPN, enhancing remote-access reliability by 23%, supporting 500+ users and reducing connectivity-related support tickets by 40%.
- Configured Amazon SNS topics and SQS queues to support alerting and background processing, reducing incident response latency by 22%.
- Automated AWS provisioning with Python Boto3, reducing manual recovery efforts by 39%.
- Developed APIs with Django REST Framework and FastAPI, improving workflow automation by 35%.
- Created Django fullstack applications for visualization, reducing RCA time by 27%.
- Applied Pandas to analyze AWS usage, improving forecasting accuracy by 30%.
- Piloted LLM-based anomaly detection with OpenAI SDK, reducing repeat incidents by 24%.
- Configured Amazon EKS clusters using Terraform and PowerShell scripts, increasing system reliability by 33%
- Implemented AWS WAF & Shield, reducing abnormal traffic incidents by 35%.
- Managed AWS Organizations, simplifying governance and reducing policy drift by 28%.
- Used AWS Cost Explorer and Trusted Advisor to identify idle resources and right-size instances, saving clients about 19% per year.
- Automated credential rotation with AWS Secrets Manager and Boto3 scripts, cutting audit findings by 31%.
- Implemented AWS KMS encryption for S3 buckets and RDS databases, reducing encryption gaps by 29% and improving compliance.
- Configured AWS Inspector scans and wrote PowerShell remediation playbooks, lowering repeat security incidents by roughly 25%.

Education:

Master of Science in Biochemistry
University of Douala, Douala, Cameroon **August 2000 – July 2002 Bachelor of Science in Engineering**
University of Douala, Douala, Cameroon **August 1995 – June 1999**

Certifications:

AWS Certified Cloud Practitioner **December 2023**
AWS Certified Solutions Architect – Associate **February 2024**
Additional Information: Languages: English (fluent), French (fluent)